Xingyun Mou

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EDUCATION

University of Southern California Los Angeles, CA
B.S. Candidate, Applied and Computational Mathematics *Jan 2024 - Present*
Minor in Artificial Intelligence Applications | **GPA: 3.93/4.0**
Academic Honors: USC Dornsife Dean's List in both 2024 and 2025
Coursework: AI, Machine Learning, Probability Theory, OOP, Discrete Methods, Algorithms & Computation Theory

University of California, Irvine Irvine, CA
Undergraduate Studies, Environmental Science and Policy *Sep 2022 - Dec 2023*
Double major in Urban Studies | **GPA: 4.0/4.0**
Academic Honors: Dean's Honor List in both 2022 and 2023
Quantitative Coursework: Linear Algebra, Basic Statistics, Calculus

RESEARCH EXPERIENCE

World Model Inference and 3D Deformation Analysis Los Angeles, CA
Undergraduate Research Assistant | University of Southern California *Feb 2026 - Present*
- Applied **model quantization** to the PSI-0.5 world model's **8B-parameter** autoregressive Transformer backbone to improve inference efficiency.
- Built an interactive **Gradio** virtual poking pipeline integrating monocular depth estimation and sparse motion prompts with PSI-predicted RGB, depth, and dense optical flow.
- Reconstructed tracked **3D surface correspondences** from world-model predictions; combined KNN neighborhood search with weighted least-squares fitting to estimate local deformation gradients.
- Computed **Green-Lagrange strain** and SVD-based principal stretches to characterize local stretch, compression, and deformation anisotropy.

SNAP Call Center Simulation and Statistical Inference Los Angeles, CA
Undergraduate Researcher | DSO Summer Scholars *May 2026 - Aug 2026*
- Developed a **Python discrete-event simulator** for Missouri SNAP call centers, modeling queueing, caller abandonment, and repeat call attempts to examine system congestion and caller outcomes.
- Validated simulation averages against **fluid-model predictions across 3 call-center settings**, comparing stochastic simulation outputs with analytical approximations.
- Compared **geometric maximum-likelihood estimates** with fluid-flow estimates of caller attempt distributions; traced a mismatch between the approaches to congestion-estimation bias.
- Tested a diagnostic queue-length substitution, reducing parameter error by **approximately 63% in that case**.
- Built a **Streamlit dashboard** for reproducible Monte Carlo experiments, parameter sweeps, and scenario comparisons of waiting times, abandonment, and caller outcomes.

LLM Mathematical Reasoning and Formal Verification Shanghai, China
Research Intern | Shanghai AI Laboratory *May 2025 - Dec 2025*
- Fine-tuned **DeepSeek-R1-Distill-Qwen-7B** with chain-of-thought supervision for multi-step mathematical reasoning, incorporating intermediate reasoning steps into supervised training.
- Built preprocessing workflows for **large-scale pretraining datasets**, covering data cleaning, normalization, and format standardization to prepare consistent training inputs.
- Explored **Lean4-based formal verification** as an approach to checking mathematical reasoning and improving the reliability of LLM-generated solutions.

Long-Term Land Cover Change and Geospatial Analysis Irvine, CA
Undergraduate Research Assistant | University of California, Irvine *Nov 2022 - Nov 2023*
- Processed **satellite imagery, geospatial layers, and time-series data** to extract structured features from large-scale datasets for environmental modeling.
- Conducted a **30-year land cover change analysis** using QGIS and remote sensing data to examine long-term spatial change.
- Used exploratory data analysis and correlation analysis to investigate nonlinear relationships and regional patterns in high-dimensional data.

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, C++, R, MATLAB | **ML Frameworks:** PyTorch, Hugging Face Transformers, scikit-learn
Applied Methods: LLM fine-tuning, model quantization, 3D reconstruction, discrete-event simulation, Monte Carlo experiments, maximum-likelihood estimation, geospatial analysis
Applications: Streamlit, Gradio, QGIS | **Development:** Git, Linux, Docker, CMake, pytest, LaTeX
Languages: Mandarin (native), English (fluent)